

CANONICAL LESSON

Samples That Can Represent a Nation

Decision-centred foundations for accountable sector intelligence

Production Unit	PU-B07-C17
Notebook	V7_A_N12
Status	Controlled educational draft v0.1
Data boundary	Synthetic instructional data only

Central decision: Design agricultural samples whose target population, frame, stratification, selection probabilities, weights, and precision support national decisions.

Learning outcomes

Why this matters

A large sample is not nationally representative when coverage, selection probabilities, or weights are defective.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

Worked cross-sector example

A stratified sample preserves selection probabilities and calculates weights, nonresponse adjustments, and design-based precision.

Boundary: The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

Practice and reflection

- 1.
- Explain why weights are required.
- 2.
- Add cluster design effects.
- 3.

- Design nonresponse adjustment.

Answer key

- 1.
- Unequal inclusion probabilities require inverse-probability weights for population inference.
- 2.
- Inflate the effective sample using justified intracluster correlation and cluster size assumptions.
- 3.
- Define response cells or a response-propensity model using pre-outcome variables, cap extreme adjustments, and validate bias/variance.

Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.